

METROPOLITAN NEUROLOGICAL INSTITUTE

Department of Neurodiagnostics & Clinical Neurophysiology

PATIENT: Jane Doe DOB: 10/14/1978 ID: NX-884920
EXAM DATE: October 08, 2026 REFERRING: Dr. S. Vance, MD
PROCEDURE: Electromyography and Nerve Conduction Study (EMG/NCS)

CLINICAL INDICATION

42-year-old computer programmer with progressive nocturnal numbness and tingling in the thumb, index, and middle fingers of the right hand. Evaluate for median neuropathy or cervical radiculopathy.

TECHNIQUE

Motor and sensory nerve conduction studies of the bilateral median and ulnar nerves were performed. Needle electromyography (EMG) was conducted in selected muscles of the right upper extremity.

FINDINGS

NERVE CONDUCTION STUDIES:

- Right Median Sensory: Distal sensory latency prolonged at 4.2 ms (normal < 3.5 ms); transcarpal sensory conduction velocity slowed.
- Right Median Motor: Distal motor latency mildly prolonged at 4.4 ms (normal < 4.0 ms) recorded at abductor pollicis brevis. Compound muscle action potential (CMAP) amplitude is preserved at 8.2 mV.
- Right Ulnar Sensory and Motor: Distal latencies, amplitudes, and conduction velocities are normal across all segments.

NEEDLE ELECTROMYOGRAPHY:

Needle electrode examination of the right abductor pollicis brevis, first dorsal interosseous, pronator teres, biceps brachii, and cervical paraspinal muscles demonstrated normal insertional activity. No resting fibrillation potentials or positive sharp waves observed. Normal motor unit action potential (MUAP) morphology and recruitment pattern.

IMPRESSION

1. Electrodiagnostic findings consistent with mild-to-moderate right median neuropathy at the carpal tunnel (Carpal Tunnel Syndrome), evidenced by sensory and motor distal latency prolongation with preserved motor amplitude.
2. No electrodiagnostic evidence of right ulnar neuropathy, plexopathy, or active cervical radiculopathy.

Electronically signed by: Marcus Sterling, MD, Neurologist / Clinical Neurophysiologist
Board Certified in Neurology and Electrodiagnostic Medicine